

1 – Introduction & Basic Concepts

- $E_{\text{in}} - E_{\text{out}} = \Delta E_{\text{sys}}$
First Law – energy conservation for a system.
- $E_{\text{total}} = U + KE + PE$, $KE = \frac{1}{2}mV^2$, $PE = mgz$
Total energy includes internal, kinetic and potential components.
- $P_{\text{abs}} = P_{\text{atm}} + P_{\text{gage}}$, $P = P_0 + \rho gh$
Absolute pressure relation and hydrostatic equation.
- $T(K) = T(^{\circ}C) + 273.15$, $T(^{\circ}F) = 1.8T(^{\circ}C) + 32$
Temperature scale conversions.
- $\rho = m/V$, $v = 1/\rho$, $\gamma = \rho g$
Density, specific volume and specific weight.
- $p_{\text{gage}} = \rho gh$, $\Delta p = (p_2 - p_1)gh$
Manometer and differential pressure relations.

2 – Energy & General Analysis

- $\Delta E = Q - W$
First Law for a closed system (energy balance).
- $\Delta U = Q - W_b - W_{\text{other}}$, $W_b = \int P \, dV$
Internal energy change including boundary work and other work modes.
- $\eta = \frac{\text{useful output}}{\text{energy input}}$
General efficiency definition.
- $w_{\text{flow}} = Pv$, $e_{\text{mech}} = Pv + \frac{V^2}{2} + gz$
Flow work and specific mechanical energy for flowing fluids.
- $e_{\text{flow}} = h + \frac{V^2}{2} + gz$, $\dot{E}_m = \dot{m}(h + \frac{V^2}{2} + gz)$
Energy transport with mass flow in open systems.

3 – Properties of Pure Substances

- $x = \frac{m_u}{m_t}$, $y = y_f + x(y_g - y_f)$
Quality and mixture property interpolation.
- $h = u + Pv$
Definition of enthalpy.
- $Pv = RT$, $Z = Pv/(RT)$
Ideal gas law and compressibility factor.
- **State Principle:** two independent intensive props fix the state.
- **Specific vs Total:** per mass properties (v, h, u, s) vs totals (V, H, U, S).

4 – Energy Analysis of Closed Systems

- $W_b = \int_{V_1}^{V_2} P \, dV$
General boundary work.
- $W_b = P(V_2 - V_1)$
Isobaric process work.
- $W_b = P_1 V_1 \ln(V_2/V_1)$
Isothermal ideal gas work.
- $W_b = \frac{P_2 V_2 - P_1 V_1}{1-n}$, $n \neq 1$
Polytropic process work.
- $W_b = \frac{P_1 + P_2}{2} (V_2 - V_1)$
Linear p-v relation (trapezoidal area).
- $Q - W = \Delta U + \Delta KE + \Delta PE$
Complete energy balance for a closed system.
- $c_v = (\partial u / \partial T)_V$, $c_p = (\partial h / \partial T)_P$
Definitions of specific heats.
- $\Delta u \approx c_v(T_2 - T_1)$, $\Delta h \approx c_p(T_2 - T_1)$
Constant-specific-heat approximation for ideal gases.
- $c_p - c_v = R$
Mayer's relation for ideal gases.

5 – Control Volumes (Mass & Energy)

- $\sum m_{\text{in}} - \sum m_{\text{out}} = \frac{dm_{\text{sys}}}{dt}$
Mass conservation for control volumes.
- $\frac{dE_{cv}}{dt} = \sum \dot{Q}_{\text{in}} - \sum \dot{Q}_{\text{out}} + \sum \dot{m} \left(h + \frac{V^2}{2} + gz \right)_{\text{in}} - \sum \dot{m} \left(h + \frac{V^2}{2} + gz \right)_{\text{out}}$
General energy rate balance for open systems.
- **Steady-Flow Devices:**
 - Nozzle/Diffuser: $h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$
Converts enthalpy kinetic energy (adiabatic).
 - Turbine: $w_t = h_1 - h_2$
Work output from enthalpy drop.
 - Compressor/Pump: $w_c = h_2 - h_1$
Work input to raise enthalpy or pressure.
 - Throttle: $h_2 \approx h_1$
Isenthalpic pressure drop (no work).
 - Mixing chamber: $\sum \dot{m}_{\text{in}} h_{\text{in}} = \sum \dot{m}_{\text{out}} h_{\text{out}}$
Adiabatic mixing, no shaft work.
 - Heat exchanger: $\dot{Q}_{\text{hot}} = -\dot{Q}_{\text{cold}}$
Energy transfer between streams (no work).

6 – Thermodynamic Property Relations

- $H = U + PV$, $h = u + Pv$
Definition of enthalpy (total and specific).
- $Tds = du + Pdv$, $Tds = dh - vdp$
Tds relations linking state properties.
- $ds = \frac{du}{T} + \frac{Pdv}{T}$, $ds = \frac{dh}{T} - \frac{vdp}{T}$
Differential forms for calculating entropy changes.
- $R = C_p - C_v$, $k = C_p/C_v$
Relations between specific heats and gas constant.
- Incompressible: $ds = c \frac{dT}{T}$, $s_2 - s_1 = c_{\text{avg}} \ln(T_2/T_1)$
Entropy change for liquids/solids (constant c).
- $dh = C_p \, dT$, $du = C_v \, dT$
Constant-property relations for ideal gases.
- $\Delta h = C_p(T_2 - T_1)$, $\Delta u = C_v(T_2 - T_1)$
Ideal-gas property change formulas.
- $\Delta s = C_v \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1}$
Entropy change in terms of T and v.
- $\Delta s = C_p \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1}$
Entropy change in terms of T and P.
- $h = u + Pv$, $du = Tds - Pdv$, $dh = Tds + vdp$
Combined property differentials for analysis of reversible paths.

7 – Entropy Fundamentals

- $\oint \frac{\delta Q}{T} \leq 0$
Clausius inequality – equality for reversible cycles.
- $dS = \left(\frac{\delta Q}{T} \right)_{\text{rev}}$
Definition of differential entropy change.
- $\Delta S = \int_1^2 \frac{\delta Q_{\text{rev}}}{T}$
Entropy change between two states.
- $\Delta S_{\text{isoth}} = \frac{Q}{T_0}$
Entropy change for reversible isothermal process.
- $\Delta S_{\text{sys}} = \int \frac{\delta Q}{T} + S_{\text{gen}}$
General form with entropy generation ($S_{\text{gen}} \geq 0$).
- $\Delta S_{\text{isol}} \geq 0$
Entropy of isolated systems never decreases.
- $Tds = dU + PdV$, $Tds = dh - vdp$
Combined 1st + 2nd law differentials.
- $S = k \ln W$, $S = -k \sum_i p_i \ln p_i$
Statistical definition of entropy.

8 – Isentropic & Reversible Processes

- $PV^k = \text{const}$ (for ideal gas)
Isentropic relation for reversible adiabatic processes.
- $T_2/T_1 = (P_2/P_1)^{(k-1)/k}$
Temperature-pressure relation, isentropic ideal gas.
- $T_2/T_1 = (v_1/v_2)^{k-1}$
Temperature-volume relation, isentropic ideal gas.
- $P_2/P_1 = (v_1/v_2)^k$
Pressure-volume form of isentropic law.
- $w = \frac{P_2 v_2 - P_1 v_1}{1-k}$
Work done in an isentropic process ($k \neq 1$).
- $q_{\text{rev}} = \int T \, ds$
Heat transfer for reversible process.

9 – Gas Power Cycles

- $\eta = 1 - \frac{Q_{\text{out}}}{Q_{\text{in}}}$
Thermal efficiency of any cycle.
- $W_{\text{net}} = Q_{\text{in}} - Q_{\text{out}}$
Net work per cycle.
- **Otto:** $\eta = 1 - \frac{1}{r^{k-1}}$, $r = V_1/V_2$
Efficiency for spark-ignition (constant-volume) engine.
- **Diesel:** $\eta = 1 - \frac{1}{r^{k-1}} \frac{(\rho^k - 1)}{k(\rho - 1)}$
Efficiency for compression-ignition (constant-pressure) engine.
- **Brayton:** $\eta = 1 - \frac{1}{r_p^{(k-1)/k}}$, $r_p = \frac{P_2/P_1}{P_1/P_1}$
Ideal gas-turbine efficiency.
- $W_{\text{net}} = c_p[(T_3 - T_4) - (T_2 - T_1)]$
Net work per unit mass for Brayton cycle.
- $\epsilon = (T_5 - T_2)/(T_4 - T_2)$
Regenerator effectiveness.
- $\eta_c = \frac{h_2s - h_1}{h_2 - h_1}$, $\eta_t = \frac{h_3 - h_4}{h_3 - h_4s}$
Isentropic efficiencies of compressor and turbine.

10 – Vapour Power Cycles (Rankine)

- $\eta = 1 - \frac{h_4 - h_1}{h_3 - h_2}$
Thermal efficiency of Rankine cycle.
- $w_t = h_3 - h_4$, $w_p = h_2 - h_1$
Turbine output and pump input per kg.
- $q_{\text{in}} = h_3 - h_2$, $q_{\text{out}} = h_4 - h_1$
Boiler and condenser heat transfers.
- $w_{\text{net}} = (h_3 - h_4) - (h_2 - h_1)$
Net cycle work.
- **Reheat:** $w_t = (h_3 - h_4) + (h_5 - h_6)$
Work for two-stage expansion with reheat.
- **Regeneration:** $y = \frac{h_f - h_2}{h_y - h_2}$
Extraction (bleed) steam fraction for open feedwater heater.
- $\text{UF} = \frac{W_{\text{net}} + Q_{\text{proc}}}{Q_{\text{in}}}$
Utilisation factor for cogeneration.

11 – Refrigeration & Heat Pump Cycles

- $\text{COP}_R = \frac{Q_L}{W_{\text{in}}}$, $\text{COP}_{HP} = \frac{Q_H}{W_{\text{in}}} = 1 + \text{COP}_R$
Definitions of coefficient of performance.
- $\text{COP}_{R,C} = \frac{T_L}{T_H - T_L}$
Carnot refrigeration COP.
- $\text{COP}_R = \frac{h_1 - h_4}{h_2 - h_1}$
Vapour-compression COP (actual).
- $w_c = h_2 - h_1$, $q_L = h_1 - h_4$, $q_H = h_2 - h_3$
Compressor work and heat transfers.
- $\eta_c = \frac{h_2s - h_1}{h_2 - h_1}$
Isentropic compressor efficiency.
- $\text{COP}_R = 1/(\frac{h_2 - h_1}{h_1 - h_4})^{(k-1)/k} - 1$
Gas-refrigeration (Brayton) COP.

12 – Isentropic Devices & Analysis

- $\Delta h = c_p(T_2 - T_1)$, $\Delta u = c_v(T_2 - T_1)$
Constant specific heat approximations for ideal gases.
- $\Delta h = \int c_p(T) \, dT$, $\Delta u = \int c_v(T) \, dT$
Variable specific heat integration forms.
- $\Delta s = \int \frac{c_p(T)}{T} \, dT - R \ln \frac{p_2}{p_1}$
Entropy change for ideal gas with variable c_p .
- $\frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right)^{(k-1)/k}$, $\frac{p_2}{p_1} = \left(\frac{v_1}{v_2} \right)^k$, $\frac{T_2}{T_1} = \left(\frac{v_1}{v_2} \right)^{k-1}$
Isentropic relations for ideal gas with constant k.
- $p_2/p_1 = P_{r2}/P_{r1}$, $v_2/v_1 = V_{r2}/V_{r1}$
Relative pressure/volume for variable c_p (use Table A-17).
- $Q - W = \Delta U + \Delta KE + \Delta PE$ (closed)
First law for closed systems.
- $\dot{Q} - \dot{W} = \dot{m}[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2}] + g(z_2 - z_1)$ (open)
First law for steady-flow devices.
- $\dot{W}_t = \dot{m}(h_1 - h_2)$
Turbine work output.
- $\dot{W}_c = \dot{m}(h_2 - h_1) \approx \dot{m}v\Delta p$
Compressor/pump work input.
- $\eta_t = \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$, $\eta_c = \frac{h_{2s} - h_1}{h_{2a} - h_1}$, $\eta_p = \frac{v_1(p_2 - p_1)}{h_{2a} - h_1}$
Isentropic efficiencies for turbine, compressor, and pump.

13 – Gas Power Cycles

- $\eta = 1 - \frac{Q_L}{Q_H} = W_{\text{net}}/Q_H$
Thermal efficiency for any heat engine cycle.
- $\text{COP}_R = \frac{Q_L}{W_{\text{net}}}$, $\text{COP}_{HP} = \frac{Q_H}{W_{\text{net}}}$
Coefficient of performance for refrigerators and heat pumps.
- $V_D = V_{\text{max}} - V_{\text{min}}$, $r = \frac{V_{BDC}}{V_{TDC}}$, $p_{\text{mep}} = W_{\text{net}}/V_D$
Displacement volume, compression ratio, and mean effective pressure.
- **Otto:** $\eta = 1 - \frac{1}{r^{k-1}}$, $w_{\text{net}} = c_v[(T_3 - T_4) - (T_2 - T_1)]$
Spark-ignition cycle efficiency and net work (1-2 isent. comp., 2-3 const-V heat add, 3-4 isent. exp., 4-1 const-V reject).
- **Diesel:** $\eta = 1 - \frac{1}{r^{k-1}} \frac{\beta^k - 1}{k(\beta - 1)}$, $\beta = V_3/V_2$
Compression-ignition cycle efficiency and cut-off ratio (2-3 const-P heat add).
- $T_2 = T_1 r^{k-1}$, $T_3 = T_2 \beta$, $T_4 = T_3(\beta/r)^{k-1}$
Temperature relations for Diesel cycle.
- **Stirling/Ericsson:** $\eta = 1 - \frac{T_L}{T_H}$
Carnot efficiency for constant-temperature heat addition/rejection cycles.
- **Brayton:** $\eta = 1 - \frac{1}{r_p^{(k-1)/k}}$, $r_p = \frac{P_{\text{max}}/P_{\text{min}}}{P_1/P_1}$
Gas-turbine cycle efficiency and pressure ratio (1-2 isent. comp., 2-3 const-P heat add, 3-4 isent. exp., 4-1 const-P reject).
- $T_{2a} = T_1 + \frac{T_{2s} - T_1}{\eta_c}$, $T_{4a} = T_3 - \eta_t(T_3 - T_{4s})$
Non-ideal Brayton temperatures with compressor/turbine efficiencies.
- $\epsilon = \frac{h_5 - h_2}{h_4 - h_2}$ (regeneration), $r_{\text{stage}} = \sqrt{r_p}$ (intercooling)
Regenerator effectiveness and optimal stage pressure ratio for Brayton improvements.
- $R_{BW} = w_c/w_t \approx 0.4-0.6$
Back work ratio for gas-turbine cycles.
- **Rankine:** $w_p = v(p_2 - p_1)$, $w_t = h_3 - h_4$, $\eta = 1 - \frac{q_{\text{out}}}{q_{\text{in}}}$
Steam cycle: pump work, turbine work, and efficiency (1-2 pump, 2-3 boiler, 3-4 turbine, 4-1 condenser).
- **Reheat:** $q_{\text{in}} = (h_3 - h_2) + (h_5 - h_4)$, $w_t = (h_3 - h_4) + (h_5 - h_6)$
Two-turbine Rankine with reheater; improves efficiency and steam quality.
- $\eta_p = \frac{v(p_2 - p_1)}{h_{2a} - h_1}$, $\eta_t = \frac{h_3 - h_{4a}}{h_3 - h_{4s}}$
Pump and turbine isentropic efficiencies for real Rankine analysis.

Isentropic Devices & Process Representations

Turbine (Expansion Device)

- **Purpose:** Converts fluid enthalpy to shaft work output.
- **Steady-flow energy:** $\dot{W}_t = \dot{m}(h_1 - h_2)$
- **Isentropic efficiency:**
$$\eta_t = \frac{h_1 - h_{2a}}{h_1 - h_{2s}}$$

h_{2s} is isentropic (ideal) exit enthalpy, h_{2a} is actual.

- **Diagrams:**
 - On $T-s$: vertical (isentropic) drop ideally; real line bends right (entropy $\uparrow$).
 - On $p-v$: steep expansion curve; area under curve $\approx$ work.
- **Notes:** Adiabatic (usually), so $Q \approx 0$ and $s_2 \geq s_1$. High-pressure vapour or gas enters, expands through blades $\rightarrow$ kinetic and shaft work output.

Compressor / Pump

- **Purpose:** Raises pressure of gas (compressor) or liquid (pump) using shaft work input.
- **Energy relation:** $\dot{W}_c = \dot{m}(h_2 - h_1)$
- **Isentropic efficiency:**
$$\eta_c = \frac{h_{2s} - h_1}{h_{2a} - h_1}, \quad \eta_p = \frac{v(p_2 - p_1)}{h_{2a} - h_1}$$

For pumps, v is nearly constant (incompressible).
- **Diagrams:**
 - On $T-s$: vertical rise ideally; real line bends right (entropy $\uparrow$).
 - On $p-v$: compression curve sloping upward; smaller area under curve = work input.
- **Notes:** Ideal isentropic $\rightarrow$ minimum work input. Real $\rightarrow$ higher T_2 , $s_2 > s_1$. Often assume adiabatic ($Q \approx 0$).

Nozzle & Diffuser

- **Purpose:** Convert enthalpy kinetic energy with negligible heat/work exchange.
- **Energy:** $h_1 + \frac{V_1^2}{2} = h_2 + \frac{V_2^2}{2}$, neglecting Δz .
- **Nozzle:** $V_2 > V_1$, $h_2 < h_1$, pressure drops, $\dot{W} = 0$.
- **Diffuser:** $V_2 < V_1$, $h_2 > h_1$, pressure rises.
- **Diagrams:**
 - $T-s$: nearly vertical line (small entropy change), slight T drop for nozzle.
 - $p-v$: nozzle expansion $\rightarrow$ down/right curve; diffuser $\rightarrow$ reverse.
- **Key parameters:** $M = V/a$ (Mach number), adiabatic $\rightarrow \Delta s \approx 0$.

Throttling (Expansion Valve)

- **Purpose:** Reduce pressure without work or heat transfer.
- **Relation:** $h_1 = h_2$ (isenthalpic).
- **Used in:** Refrigeration and Rankine regeneration systems.
- **Diagrams:**
 - $T-s$: sharp horizontal line (constant h , entropy increases).
 - $p-v$: steep expansion curve.
- **Notes:** Always irreversible; causes T drop in two-phase or gas with Joule–Thomson coefficient $\mu_{JT} > 0$.

Heat Exchanger (Boiler, Condenser, Regenerator)

- **Purpose:** Transfer heat between two fluids with no work.
- **Relation:**
$$-Q_{cold} \sum \dot{m} h_{in} = \sum \dot{m} h_{out}$$
- **Diagrams:**
 - $T-s$: parallel constant-pressure lines for hot/cold streams.
 - $p-v$: little movement (pressure constant).
- **Notes:** Used in cycles as boiler (adds heat at high P), condenser (removes heat at low P), regenerator (recovers waste heat).

Mixing & Separation Chambers

- **Purpose:** Combine or split mass flows with negligible heat/work.
- **Energy balance:** $\sum \dot{m}_{in} h_{in} = \sum \dot{m}_{out} h_{out}$.
- **Diagrams:** No clear single path — represent as vertical s -weighted mixing on $T-s$.
- **Notes:** Adiabatic mixing increases entropy ($s_{mix} > s_{in}$). Used in open feedwater heaters, combustion chambers.

How to Draw Processes on Diagrams

p–v Diagram

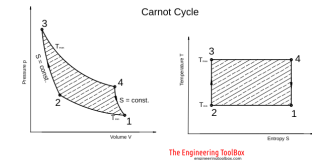
- p on vertical axis (log scale often), v horizontal.
- Area under curve $\Rightarrow$ boundary work ($W_b = \int P dV$).
- Steeper slopes $\rightarrow$ lower compressibility.
- Common shapes:
 - Isothermal: rectangular hyperbola ($PV = \text{const}$)
 - Adiabatic/Isentropic: steeper than isothermal ($PV^k = \text{const}$)
 - Isobaric: horizontal line
 - Isochoric: vertical line

T–s Diagram

- T on vertical, s horizontal. Area under curve = heat transfer $q_{rev} = \int T ds$.
- **Isentropic:** vertical line ($\Delta s = 0$)
- **Isothermal:** horizontal line
- **Isobaric:** curve leaning right (for heating) or left (for cooling)
- **Two-phase region:** horizontal lines (constant T , P)
- Cycle efficiency visualised by enclosed area (work output/input).

Major Cycle Diagrams & Efficiencies

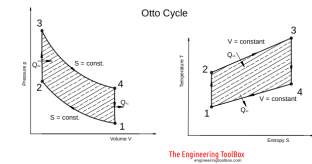
Carnot Cycle¹



$$\eta_{\text{Carnot}} = 1 - \frac{T_{\min}}{T_{\max}}$$

$1 \rightarrow 2$ isothermal expansion, $2 \rightarrow 3$ isentropic expansion, $3 \rightarrow 4$ isothermal compression, $4 \rightarrow 1$ isentropic compression.

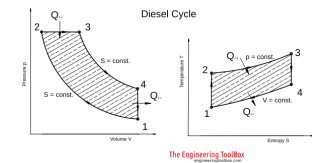
Otto Cycle²



$$\eta_{\text{Otto}} = 1 - \frac{1}{r^{k-1}}$$

$r = V_1/V_2$ compression ratio. $1 \rightarrow 2$ isentropic comp., $2 \rightarrow 3$ const- V heat add, $3 \rightarrow 4$ isentropic exp., $4 \rightarrow 1$ const- V reject.

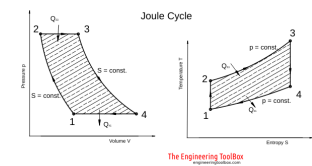
Diesel Cycle³



$$\eta_{\text{Diesel}} = 1 - \frac{1}{r^{k-1}} \cdot \frac{\beta^k - 1}{k(\beta - 1)}$$

$\beta = V_3/V_2$ cutoff ratio. $2 \rightarrow 3$ is constant-pressure heat addition (fuel injection).

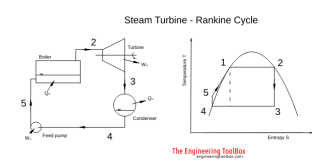
Joule (Brayton) Cycle⁴



$$\eta_{\text{Joule}} = 1 - \frac{T_1}{T_2} = 1 - \frac{1}{r_p^{(k-1)/k}}$$

$r_p = P_2/P_1$ pressure ratio. Used in gas turbines with continuous flow.

Rankine Cycle (Steam)⁵



$$\eta_{\text{Rankine}} = 1 - \frac{h_4 - h_1}{h_3 - h_2}$$

$w_t = h_3 - h_4$ (turbine), $w_p = h_2 - h_1$ (pump). $1 \rightarrow 2$ pump, $2 \rightarrow 3$ boiler, $3 \rightarrow 4$ turbine, $4 \rightarrow 1$ condenser.

Cross-Links to Key Equations

- **Energy Balance:** $Q - W = \Delta U + \Delta KE + \Delta PE$ (Sec. 4)
- **Steady-flow:** $\dot{Q} - \dot{W} = \dot{m}[(h_2 - h_1) + \frac{V_2^2 - V_1^2}{2} + g(z_2 - z_1)]$ (Sec. 5)
- **Entropy:** $T ds = dh - v dP$ (Sec. 6–7)
- **Isentropic Relations:** $T_2/T_1 = (P_2/P_1)^{(k-1)/k}$ (Sec. 8–12)
- **Efficiencies:**

$$\eta_t = \frac{h_1 - h_{2a}}{h_1 - h_{2s}}, \quad \eta_c = \frac{h_{2s} - h_1}{h_{2a} - h_1}, \quad \eta_p = \dots$$

¹Carnot: Ideal reversible cycle; maximum possible efficiency between $T_{\max}$ and $T_{\min}$. Impractical due to isothermal process difficulty.

²Otto: Spark-ignition engines (gasoline). Typical $r = 8\text{--}12$. Higher compression $\rightarrow$ higher efficiency.

³Diesel: Compression-ignition engines. Higher compression ratios (14–24). When $\beta = 1$, equals Otto efficiency.

⁴Joule/Brayton: Gas turbines, jet engines. Efficiency $\uparrow$ with pressure ratio. Improvements: regeneration, intercooling, reheat.

⁵Rankine: Steam power plants. Reheat and regeneration improve efficiency. Pump work negligible compared to turbine work.